Manan Kumar

AI Engineer · LLM Agents · RAG · Evals · Voice AI · Python / FastAPI

New Delhi, India (open to remote & relocation) | 04manank@gmail.com | +91 88511 73040

linkedin.com/in/manan-kumar-2229291b9 | github.com/Manan0802

SUMMARY

AI Engineer shipping production generative AI in Python/FastAPI: tool-calling LLM agents, RAG, evals and real-time voice. Top contributor to a fintech startup's Hinglish (Hindi-English) AI assistant for retail savers, where I wrote most of the agent code, raised intent accuracy from 68% to 96% and cut model calls per reply from 3–8 to 1–2.

EXPERIENCE

Software Development Engineer I (AI) | **Fintech Startup** Jun 2026 - Present

Hinglish AI assistant for retail savers · Python, FastAPI, PostgreSQL + pgvector, Redis, Gemini, AWS Bedrock

- **Raised intent-routing accuracy from 68% to 96%** on a frozen 193-case held-out Hinglish set by replacing a rule-based router with an LLM classifier, cutting model calls per reply from 3–8 to 1–2.
- **Shipped a tool-calling RAG agent to production in app chat** via a percentage A/B rollout with a kill switch: consolidated 43 narrow tools into 19, lifting first-tool accuracy from 73% to ~80% on a 314-turn labeled set.
- **Built the safety layer for regulated fintech:** a money guard that keeps the model off transaction APIs, PAN (tax ID) and phone masking on model calls, and a scope rule that cut answered off-topic probes from 15 to 0 in live tests.
- **Built in-chat money actions:** changes to SIPs (recurring investments), investments and withdrawals behind consent, OTP and UPI steps, with at-most-once writes (atomic Redis claim) and money code the model cannot reach; 254 tests.
- **Built eval harnesses and cost tracking:** a blind routing gate with paired significance tests, tool-choice evals on golden sets, production replay scored by an LLM judge, p50/p95 latency evals, and per-call USD cost in Langfuse.
- **Hardened real-time voice** over WebSockets (co-built): fixed production interruption handling (barge-in) by sharing voice profiles across pods via Redis, and cut TTS failover delay after a hung connection from 10 s to 0.5 s.

Associate Software Engineering Intern – AI Engineer | **IndiaMART InterMESH Ltd.** Jan 2026 - Jun 2026

Agentic AI for B2B product-catalog quality · Python, LangGraph, Gemini 2.5 Pro, Langfuse, GCP

- **Built SARA, an autonomous spec-auditing agent run on IndiaMART's live catalog:** a LangGraph state machine with 10 on-demand skills (domain expert, critic, validator) and a validation gate that loops failed output back for correction.
- **Built a knowledge compiler used by catalog teams** that turns buyer calls, seller PDFs and web search into 14-section category wikis, updating only changed sources, rolling back score regressions and flagging open doubts for review.
- **Built a LangGraph spec-correction pipeline** merging 6 data sources through rule-based quality gates, and a serverless Google Cloud Function that pushes normalized specs to Kafka via a publish API, with GCS audit snapshots.

PROJECTS

ShopLens, visual fashion search engine | Python, CLIP, FAISS, YOLO, Streamlit

- Multimodal search over **45,000+ products**: CLIP embeds image and text queries into one vector space, FAISS retrieves matches, and color-aware re-ranking refines them. Code: github.com/Manan0802/shoplens-updated

InvestMate, AI portfolio tracker | React, Node.js, Express, MongoDB, Gemini API

- Full-stack app with JWT auth that tracks stocks and crypto with quotes from 3 market APIs and generates a structured Gemini risk summary of the user's holdings. Code: github.com/Manan0802/investmate

SKILLS

AI / LLM: Generative AI, LLM agents, tool calling, RAG, LangGraph, multi-agent systems, prompt and context engineering, prompt caching, evals (LLM-as-judge), guardrails, structured outputs, agent memory, vector databases (pgvector, FAISS)

Models & Voice: Gemini, AWS Bedrock, OpenAI, Sarvam AI, streaming speech-to-text and text-to-speech, CLIP, YOLO

Backend & Data: Python, FastAPI, SQLAlchemy, REST APIs, WebSockets, PostgreSQL, Redis, BigQuery

Tools & Other: GCP, Langfuse, Git, pytest, Claude Code · TypeScript, JavaScript, SQL, Java (Spring Boot), Kotlin, React

EDUCATION

B.Tech, Software Engineering | Delhi Technological University (DTU), New Delhi | CGPA 7.71/10 2022 - 2026

ACHIEVEMENTS

LeetCode: 400+ problems solved, 1600+ contest rating | **Coursera:** Machine Learning (Stanford), RAG & Agentic AI (IBM)